

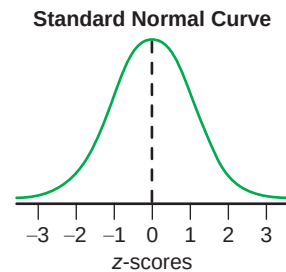
The Standard Normal Curve

The standard normal curve is used to analyze data.

When all of the data values of a normal distribution are turned into z -scores, the resulting distribution is called the standard normal curve.

Each value on the horizontal axis of the standard normal curve is a z -score, and the mean is always 0.

For instance, a data value with a z -score of 0.54 would correspond to 0.54 on the horizontal axis of the standard normal curve.



Properties of the Standard Normal Curve

The **standard normal curve** is a probability distribution with the following properties:

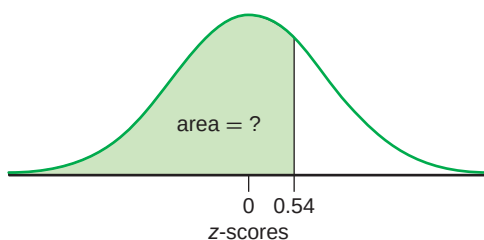
- The mean is 0.
- The standard deviation is 1.
- The total area under the curve is 1.

Using the z -Distribution Table

Because the standard normal curve is a probability distribution, it can be used to find probabilities. These probabilities correspond to areas under the standard normal curve.

Suppose a data value has a z -score of 0.54. What percent of the data values are less than this value?

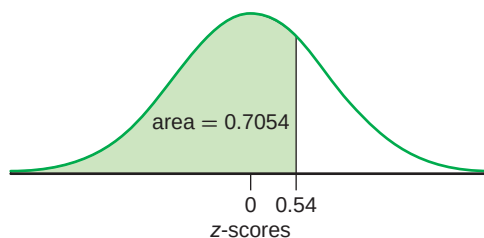
► **Remember** If a z-score is positive, then it is above the mean. If a z-score is negative, then it is below the mean.



To answer this question, you can use the Table of z-Scores for Normal Distribution in the appendix. This table gives the areas under the standard normal curve that are less than a given z-score.

<i>z</i>	.00	.01	.02	.03	.04	.05
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422

Using the z-distribution table, look for the area that corresponds to the row and column of a z-score of 0.54. You can see that $P(z < 0.54) = 0.7054$, which is about 71%.



So about 71% of the data values are less than a value with a z-score of 0.54.

► **Think About It** Technology, such as calculators and computers, can also be used to find areas under the standard normal curve.

Using the Standard Normal Curve to Solve Problems

Suppose scores on a standardized test taken by future doctors are normally distributed with a mean of 10.5 and a standard deviation of 2.2. If 150 future doctors take the test, how many are likely to score between 12 and 15?

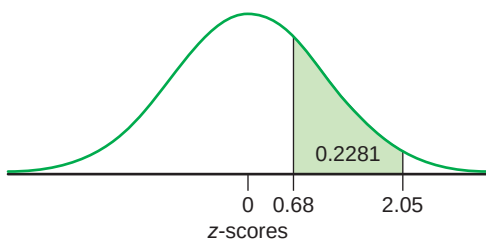
► **Remember** z-scores are obtained using the following formula:

$$z = \frac{x - \mu}{\sigma}$$

The score 12 has a z-score of about 0.68, and the score 15 has a z-score of about 2.05. To help answer the question, you need to know the area that corresponds to $P(0.68 < z < 2.05)$.

The area under the standard normal curve less than $z = 0.68$ is 0.7517, and the area less than $z = 2.05$ is 0.9798.

So $P(0.68 < z < 2.05) = 0.9798 - 0.7517 = 0.2281$.



Multiply the area by the number of future doctors taking the exam.

$$0.2281 \cdot 150 \approx 34$$

So about 34 of the 150 future doctors taking the exam are likely to score between 12 and 15.